

Chapter 5: Deformation and Strain

Reading: [Fung/94] Chapter 5

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Contents

(§5.1) Deformation

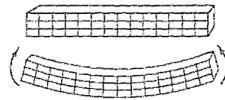
Forces applied to solids cause deformation, and forces applied to liquids cause flow. Often, the major objective of an analysis is to find the deformation or flow. It is our objective in this chapter to analyze the deformation of solid bodies in such a way as to be relevant to the state of stress in those bodies.

Questions

- How can you picture/visualize deformation?
 - Scribing regularly spaced lines on a slender rod/bar or a rubber band and axially loading it in tension so as to induce a 2% deformation?
 - * 10% strain? 100% strain??
 - * the notion of relative deformation, leading to the definition of *strain*, ϵ_a .
 - What happens when the axial load on the bar is reversed, what is the expected deformation mode(s) in compression?
 - * Can the relation of the applied load versus axial displacement of the bar be predicted for a stiff bar?

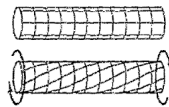
Bending

- A bar with a rectangular inscribed grid on it, bent by an equal and opposite pair of transverse couple moments at its two ends. What is the expected configurational change?
 - The initially straight longitudinal axis deforms into a nearly circular arc shape?
 - Transverse cross-sections remain plane?
 - Longitudinal fibers below the centroidal plane elongate, while the fibers above contract.



Twisting

- A similar bar is twisted by an equal and opposite pair of axial couple moments at its two ends. How is it expected to deform?
 - The longitudinal axis stays straight, with a slight amount of contraction (shortening).
 - The transverse planes remain plane through the process.
 - An angle of twist results from the accumulation of relative rotation of sections from one end to the other.



Definitions

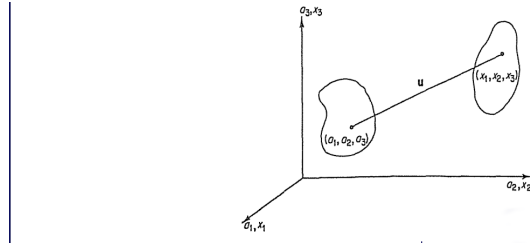
Configurations (undeformed, *initial*, reference, etc. and deformed, *current*, etc.)

- Concepts related with deformation
 - Initial length, L_0 .
 - Deformed length, L .

- Stretch ratio, L/L_0 .
- Strain⁽¹⁾, $\epsilon = \frac{L-L_0}{L_0}$ or $\epsilon' = \frac{L-L_0}{L}$
- These are also known as the *material* and *spatial* strain measures, respectively.
- Strain⁽²⁾, $E/ = \frac{L^2-L_0^2}{L_0^2}$ or $e/ = \frac{L^2-L_0^2}{L^2}$
- ⁽¹⁾ is known as *engineering* strain measure, while ⁽²⁾ is known as the *finite* strain measure.

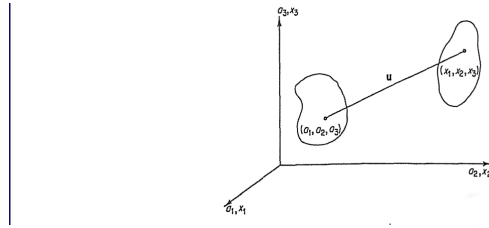
Query AI:

How can the difference between the configurational notions represented by the words /initial/ and /current/ be explained to an undergraduate biomedical engineering program cohort?



Deformation as a Process from Initial to Current

- We use the points of a body, $\mathcal{B}$, to describe the deformation process between the initial configuration (coordinates a_1, a_2, a_3) and current configuration (coordinates x_1, x_2, x_3).



- Note the duality used in this description, the notions of “initial” and “current”.
- The motion of a point, P , from its initial configuration to its current configuration, P' , is called its *displacement*, $\mathbf{u} = \overline{PP'}$.

Mapping Functions

Mathematical definition of the body's motion from initial to current state.

- The deformation process can be simulated by mapping functions which take the initial coordinates, a_i , as arguments and yield the current coordinates, x_i , as output

$$x_i = \chi_i(a_1, a_2, a_3)$$

- There has to be a unique *inverse* mapping of this motion, given by functions which take the current coordinates as arguments and yield the initial coordinates as output

$$a_i = \chi_i^{-1}(x_1, x_2, x_3)$$

- $\chi(a_1, a_2, a_3)$ and $\chi_i^{-1}(x_1, x_2, x_3)$ can be called as the *forward* and *backward* mapping functions, respectively.

Displacement of a Point

- During deformation a point P initially at (a_1, a_2, a_3) moves to P' at (x_1, x_2, x_3) . The *displacement* of P can be obtained in terms of the initial coordinates, by subtracting the initial coordinates from the current ones as

$$u_i(a_1, a_2, a_3) = \chi_i(a_1, a_2, a_3) - a_i$$

- Alternatively, the displacement can also be computed in terms of the current coordinates as

$$u_i(x_1, x_2, x_3) = x_i - \chi_i^{-1}(x_1, x_2, x_3)$$

- In this course, the former definition will be used.

Examples

5.9

Consider the square plate of unit size shown below. Find the forward mapping functions from a_i to x_i ?

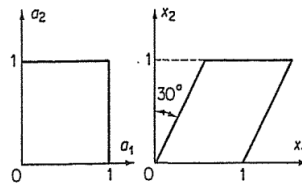
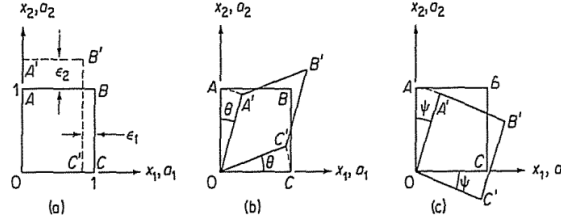


Figure P5.9 Deformation of a square plate.

5.12

A unit square $OABC$ is distorted to $OA'B'C'$ in three ways as shown below. In each of the cases write down the displacement field u_1, u_2 of every point in the square as a function of the location (a_1, a_2) .



Python Snippets

wavy mapping function

a Python script using Matplotlib to visualize a trigonometric deformation mapping. It deforms a regular grid using a wavy, trigonometric function.

```
import numpy as np
import matplotlib.pyplot as plt

def plot_deformed_grid(map_func, xlim=(-3, 3), ylim=(-3, 3), grid_size=20):
    """
    Plots a regular grid and its deformation by map_func.
    """
    # Create the original grid
    x = np.linspace(xlim[0], xlim[1], grid_size)
    y = np.linspace(ylim[0], ylim[1], grid_size)
    X, Y = np.meshgrid(x, y)

    # Apply the deformation function
    X_deformed, Y_deformed = map_func(X, Y)

    # Plot
    fig, ax = plt.subplots(figsize=(8, 8))
    # Plot original grid in light grey
    for i in range(grid_size):
        ax.plot(X[i, :], Y[i, :], color='lightgrey', linewidth=1)
        ax.plot(X[:, i], Y[:, i], color='lightgrey', linewidth=1)
```

```

# Plot deformed grid in blue
for i in range(grid_size):
    ax.plot(X_deformed[i, :], Y_deformed[i, :], color='C0', linewidth=1)
    ax.plot(X_deformed[:, i], Y_deformed[:, i], color='C0', linewidth=1)
ax.set_xlim(xlim)
ax.set_ylim(ylim)
ax.set_aspect('equal')
ax.axis('off')
plt.show()

# Define a trigonometric deformation function
def wavy_deformation(x, y):
    return x + 0.5 * np.sin(2 * y), y + 0.5 * np.cos(2 * x)

# Generate the plot
# plot_deformed_grid(wavy_deformation)

## how to run
# start a python REPL in emacs (M-x run-python)
# from deformation_wavy import plot_deformed_grid, wavy_deformation

```

Consider using <https://www.pythonanywhere.com/login/> to give this code a try!

shear animation

a Python script that creates a time-dependent simulation of a simple homogeneous deformation (pure shear) and saves it as a GIF.

```

import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
from matplotlib.patches import Circle

# Create the original grid
x = np.linspace(-2, 2, 9)
y = np.linspace(-2, 2, 9)
X, Y = np.meshgrid(x, y)

# Set up the figure and axis
fig, ax = plt.subplots(figsize=(8, 8))

```

```

ax.set_xlim(-4, 4)
ax.set_ylim(-4, 4)
ax.set_aspect('equal')
ax.axis('off')
lines = []

# Create the initial grid lines
for i in range(len(x)):
    line, = ax.plot(X[i, :], Y[i, :], 'lightgrey', lw=1)
    lines.append(line)
    line, = ax.plot(X[:, i], Y[:, i], 'lightgrey', lw=1)
    lines.append(line)

# Add a small circle to visualize local deformation
circle = Circle((0, 0), 0.2, fill=False, color='red', lw=2)
ax.add_patch(circle)

# Update function for animation
def update(frame):
    t = frame / 10 # Time parameter from 0 to 1
    # Pure shear deformation:  $F = \begin{bmatrix} 1 & k(t) \\ 0 & 1 \end{bmatrix}$ 
    k = 2 * t # Shear magnitude increases with time
    X_deformed = X + k * Y
    Y_deformed = Y

    # Update grid lines
    for i in range(len(x)):
        lines[i*2].set_data(X_deformed[i, :], Y_deformed[i, :])
        lines[i*2+1].set_data(X_deformed[:, i], Y_deformed[:, i])

    # Update the circle (approximated by a polygon)
    theta = np.linspace(0, 2*np.pi, 20)
    circle_x = 0.2 * np.cos(theta)
    circle_y = 0.2 * np.sin(theta)
    circle_x_deformed = circle_x + k * circle_y
    circle_y_deformed = circle_y
    circle.center = (0, 0) # Update the center position
    #circle.set_xy(np.column_stack([circle_x_deformed, circle_y_deformed]))

    return lines + [circle]

```

```
# Create the animation
anim = FuncAnimation(fig, update, frames=21, interval=200, blit=True)

# Save as a GIF (requires imagemagick or pillow)
# anim.save('time_dependent_shear.gif', writer='pillow', fps=5)

plt.show()
```

(§5.2) Strain

- What is strain? A stress-like second-order tensor which carry deformational information.
- Why is it important? While stress and strain are related to each other through material behavior, it is only possible to directly measure strain. Stress can only be inferred from a strain measurement.

Seeing Strain via Photoelasticity

- What does strain look like? Strain is a second-order tensor field. When transparent parts (like glass) are deformed, their light transmitting behaviour is effected by the strain field. As a result we capture optical fringe patterns by photography, known as photoelasticity. [1]

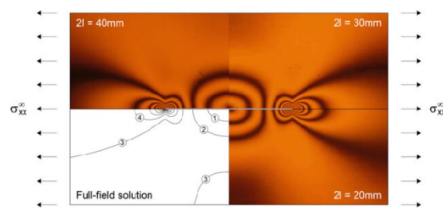


Fig. 8 Monochromatic photoelastic fringes revealing the in-plane principal stress difference field near a thin line inclusion (a 0.1 mm thick steel platelet) embedded in an elastic matrix (a two-component 'soft' epoxy resin) compared to the elastic solution (in plane strain, with Poisson's ratio equal to 0.45 and

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Measuring Strain

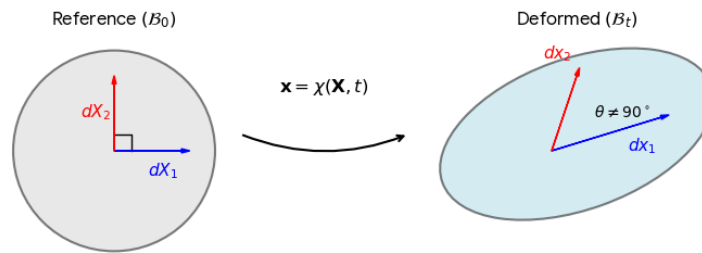
- There are various method to measure strain in a part. One of the most popular methods is by strain gage attachment on the outer surface of a part. More modern techniques involve observing parts by optical cameras and applying digital image correlation to convert the images to strain fields. [2] `./images/animation_rtaImage.gif`

Strain Types

Normal Strains

The change in length of a directed fiber, e.g., $d\mathbf{X}_1$, $d\mathbf{X}_2$

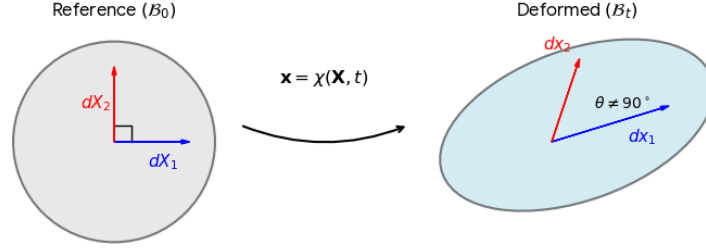
- This mode of deformation is known as dilatational, involving size changes.
- Length changes are associated with the term **normal** strain.
- In 3D space, normal strain readings need to be taken from at least three non-parallel fiber directions in order to construct the strain tensor



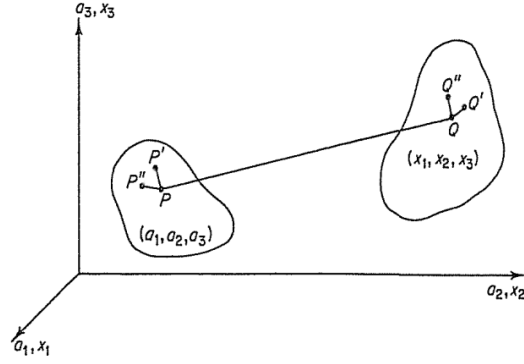
Shear Strains

The change in the angle between a pair of directed fibers.

- This mode of deformation is known as distortional, involving shape changes.
- Angle changes are associated with the term **shear** strain.
- In 3D space, shear strain readings need to be taken from at least three pairs of non-parallel fibers in order to finalize the strain tensor.



Mathematical Definition of Strain

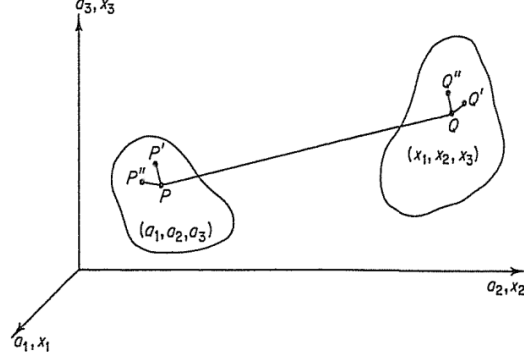


- Consider a short fiber connecting the points $P(a_1, a_2, a_3)$ and $P'(a_1 + da_1, a_2 + da_2, a_3 + da_3)$. The square of the lengths ds_0 of PP' in the initial configuration is

$$ds_0^2 = da_1^2 + da_2^2 + da_3^2$$

- As a result of the motion, PP' is moved to QQ' in the current configuration, where $Q(x_1, x_2, x_3)$ and $Q'(x_1 + dx_1, x_2 + dx_2, x_3 + dx_3)$, and the square of the length, ds of the deformed fiber QQ' is

$$ds^2 = dx_1^2 + dx_2^2 + dx_3^2$$



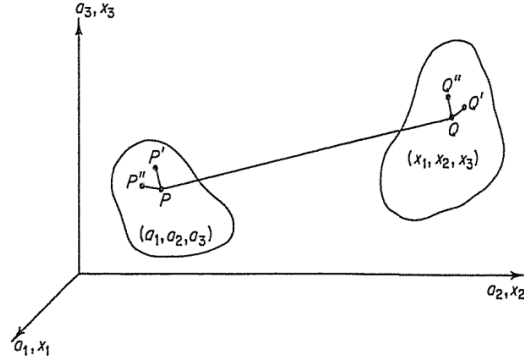
- The components of the current fiber, dx_i can be related to the components of the initial fiber da_i using the chain rule of differentiation on the mapping functions,

$$dx_i = \frac{\partial \chi_i}{\partial a_j} da_j$$

- Making use of the indicial representation of the scalar (dot) product, the square of length of the fiber element in its initial and deformed configuration can be expressed as

$$ds_0^2 = \delta_{ij} da_i da_j \quad (1)$$

$$ds^2 = \delta_{ij} dx_i dx_j = \delta_{ij} \frac{\partial \chi_i}{\partial a_l} da_l \frac{\partial \chi_j}{\partial a_m} da_m = \delta_{ij} \frac{\partial \chi_i}{\partial a_l} \frac{\partial \chi_j}{\partial a_m} da_l da_m \quad (2)$$



- The difference between the squares of the length elements may be written, after appropriate changes in the dummy indices as

$$ds^2 - ds_0^2 = \left(\delta_{lm} \frac{\partial \chi_l}{\partial a_i} \frac{\partial \chi_m}{\partial a_j} - \delta_{ij} \right) da_i da_j = 2E_{ij} da_i da_j$$

- The finite strain tensor E_{ij} introduced above is

$$E_{ij} = \frac{1}{2} \left(\delta_{lm} \frac{\partial \chi_l}{\partial a_i} \frac{\partial \chi_m}{\partial a_j} - \delta_{ij} \right)$$

- Also note that strain can be cast into a matrix form. It is symmetric. The diagonal terms are the normal strains while the off-diagonal ones are the shear strains.

Examples

5.9

Consider the square plate for which the forward mapping functions from a_i to x_i were determined. Compute the corresponding finite strain tensor.

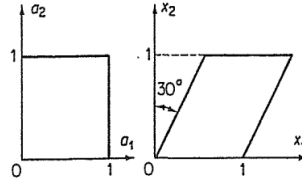
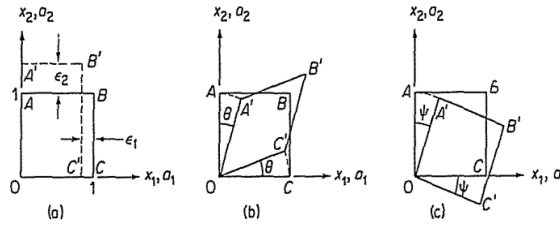


Figure P5.9 Deformation of a square plate.

5.12

Consider the square plate for which the forward mapping functions from a_i to x_i were determined for the three cases. Compute the corresponding finite strain tensor for each case.



Deformation Gradient, $\mathbf{F}$

Given the **mapping** from the initial to current configuration, $\chi_i(a_1, a_2, a_3)$, it's gradient can be taken by

$$\mathbf{F} = \nabla \chi \quad (3)$$

$$= \left(\mathbf{e}_i \frac{\partial}{\partial a_i} \right) \otimes (\chi_j \mathbf{e}_j) \quad (4)$$

$$= \frac{\partial \chi_j}{\partial a_i} \mathbf{e}_i \otimes \mathbf{e}_j \quad (5)$$

$$= \chi_{j,i} \mathbf{e}_i \otimes \mathbf{e}_j \quad (6)$$

which is known as the **deformation gradient** tensor.

Polar Decomposition

This is a process where a second-order tensor can be split into a product of two tensors, in two ways, as:

$$\mathbf{F} = \mathbf{U}\mathbf{R} = \mathbf{R}\mathbf{V}$$

where $\mathbf{R}$ is a pure rotation and $\mathbf{U}$ and $\mathbf{V}$ are known as the **right** and **left stretch** tensors.

Octave (GNU Octave / MATLAB)

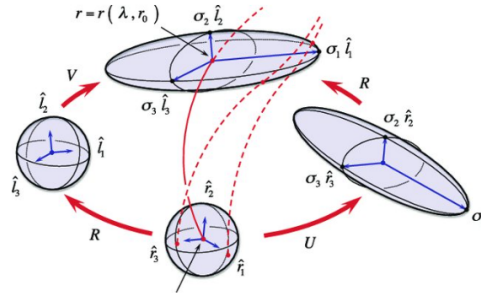
```
F = [1.2, 0.5; 0.2, 0.9];           % example deformation gradient
[U, R] = poldec(F);                 % right: F = R * U
[V, R] = poldec(F');                % left: F = V * R (using transpose trick)
V = V';                             % make V match definition
```

Python (with SciPy)

```
import numpy as np
from scipy.linalg import polar

F = np.array([[1.2, 0.5], [0.2, 0.9]]) # example F

R, U = polar(F, side='right')           # F = R @ U
R, V = polar(F, side='left')            # F = V @ R
```



Caption

Figure 5 Illustration of the polar decomposition of the Spatial Propagator. The right-polar decomposition, $F = R \cdot U$, stretches/compresses by a factor of σ_i along the corresponding direction $\hat{e}_i$, followed by a rotation aligning with $\hat{e}_i$ with α_i ; the left-polar decomposition, $F = V \cdot R$, rotates $\hat{e}_i$ rotates α_i ... [Read more](#)

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Query AI:

Provide a clear definition of the polar decomposition of the deformation gradient. If possible, provide illustrations or suggest references or sources that point graphical illustrations depicting the nature of U and V .

Coding, Analysis and Interpretation

For every mapping listed in the sections below, a complete exercise consists of the following tasks carried out in the student's programming language of choice. The tasks are listed here once and apply uniformly to all cases.

Task 1 — Visualise the deformed grid

Map a regular rectangular grid of material points (a_1, a_2) through the deformation $x_j = \chi_j(a_i)$ and plot the deformed configuration alongside the reference grid. This is the most immediate visual check: the student sees directly how straight material lines curve, rotate, stretch or fold.

Use the python snippet `./snippets/deformation.py` for visualization.

Task 2 — Compute the deformation gradient $\mathbf{F}$

Evaluate the deformation gradient component by component:

$$F_{ij} = \frac{\partial \chi_j}{\partial a_i} = \chi_{j,i}$$

For inhomogeneous mappings $\mathbf{F}$ is a function of position; verify that $J = \det(\mathbf{F}) > 0$ everywhere, or locate the points where this condition is violated and discuss their physical meaning (self-intersection, folding of the body, etc.).

Task 3 — Compute the polar decompositions of $\mathbf{F}$

Evaluate $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{R}$.

Task 4 — Compute the finite strain tensor $\mathbf{E}$

Form the previous derivation of the finite strain tensor, the finite strain tensor can be written in terms of the deformation gradient as

$$E_{ij} = \frac{1}{2}(F_{ik}F_{jk} - \delta_{ij})$$

Homogeneous Deformations

In a homogeneous deformation the deformation gradient $\mathbf{F}$ is *constant* throughout the body — every material neighbourhood deforms identically. The mapping is linear in the material coordinates a_i . These cases are the natural starting point because $\mathbf{F}$, $\mathbf{C}$ and $\mathbf{E}$ are uniform fields and can be evaluated by hand before being verified numerically.

Case 1 — Pure dilatation

$$x_1 = \lambda a_1, \quad x_2 = \lambda a_2$$

$\mathbf{F}$ is a scalar multiple of the identity. All line elements stretch equally in every direction. Determinant of the deformation gradient, $J = \lambda^2$. Because $F_{ij} = \lambda \delta_{ij}$ the tensor $C_{ij} = \lambda^2 \delta_{ij}$ and the Green–Lagrange strain $E_{ij} = \frac{1}{2}(\lambda^2 - 1)\delta_{ij}$ is diagonal with no shear components.

Physical example: a point on or in an expanding balloon.

Case 2 — Uniaxial extension with lateral contraction

$$x_1 = \lambda a_1, \quad x_2 = \mu a_2, \quad \lambda \neq \mu$$

The case $\lambda\mu = 1$ yields an isochoric (volume-preserving) deformation. This is the prototype for a uniaxial tension test.

Physical example: A rubber band stretched between two fingers. The band elongates along its axis and narrows in the transverse direction. If the rubber is nearly incompressible $\mu\lambda \approx 1$, making this also the natural place to introduce the concept of the Poisson ratio as the ratio of lateral strain response to an induced axial strain.

Case 3 — Pure shear

$$x_1 = \lambda a_1, \quad x_2 = \frac{1}{\lambda} a_2$$

Isochoric ($J = 1$) with no rotation. Often confused with simple shear by students; the comparison between Cases 3 and 4 is one of the most instructive exercises in the catalogue. The principal axes of strain coincide with the coordinate axes, whereas in simple shear they do not.

Physical example: A thin-walled cylindrical pressure vessel under combined axial compression and internal pressure, tuned so that the two stresses are equal and opposite. In the material the principal axes of strain coincide with the vessel axes, and no shear appears on those faces — a state called pure shear in the stress sense. A square element cut at 45° to the axis experiences maximum shear with no normal strain, which is what students see when they rotate the strain tensor from the principal frame.

Case 4 — Simple shear

$$x_1 = a_1 + \kappa a_2, \quad x_2 = a_2$$

κ is the shear parameter ($\kappa = \tan \theta$ for shear angle θ). The deformation is isochoric ($J = 1$) and contains both stretch and rotation. The Green–Lagrange strain components are:

$$E_{12} = E_{21} = \frac{\kappa}{2}, \quad E_{22} = \frac{\kappa^2}{2}, \quad E_{11} = E_{33} = 0$$

The diagonal component $E_{22} = \kappa^2/2$ is the key pedagogical target: it is a purely nonlinear term representing the elongation of material fibres along $\mathbf{e}_2$ caused by the shear, invisible to the linearised theory.

Physical example: A rectangular block of soft tissue or rubber bonded between two rigid plates, with one plate sliding parallel to the other — the standard lap-joint or elastomeric bearing geometry. The deformation is exactly simple shear. Engineering examples include rubber vibration isolators, bridge bearing pads, and the shear layer in a sandwich panel. The nonlinear term $E_{22} = \kappa^2/2$ has a measurable consequence in rubber: the normal stress difference that tends to push the plates apart, known as the Poynting effect.

Case 5 — General homogeneous deformation

$$x_1 = \alpha a_1 + \beta a_2, \quad x_2 = \gamma a_1 + \delta a_2$$

The deformation gradient is the 2×2 matrix $[F] = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$. All previous homogeneous cases are special instances. Students perform the polar decomposition $F = RU$ numerically, extract the rotation R and the right stretch tensor U , and connect the kinematics to the spectral theory of symmetric tensors.

Physical Example: A sheet of metal passing through a rolling mill with misaligned rolls, so that the sheet is simultaneously compressed in thickness, stretched in the rolling direction, and subjected to an asymmetric shear across its width. Every small neighbourhood of the sheet undergoes the same general linear map, making this a textbook case of homogeneous deformation at the process scale.

Case 6 — Rigid body rotation

$$x_1 = a_1 \cos \theta - a_2 \sin \theta, \quad x_2 = a_1 \sin \theta + a_2 \cos \theta$$

$F = R$ is orthogonal, so $C = F^T F = I$ and $E = 0$ exactly. This is a crucial sanity check: students verify that a pure rotation produces zero Green–Lagrange strain, confirming that E correctly filters out rigid motion. The linearised strain ε fails this test for large θ — a vivid demonstration of the restriction of the small-strain approximation to small rotations as well as small stretches.

Physical Example: A rigid flywheel or gear rotating about its axis. Every material point moves, the displacement field is non-trivial, yet no strain is developed anywhere in the body. This is the definitive physical argument for why engineering strain $\Delta L/L$ is insufficient as a strain measure for large rotations, and why a strain tensor that vanishes identically for rigid rotation — as E does — is the correct kinematic objective.

Inhomogeneous Deformations

Here $F = F(a_i)$ varies from point to point. The mappings remain analytically tractable, bridging the gap between the uniform-field homogeneous cases and full boundary-value problems.

Case 7 — Pure bending into a circular arc

$$x_1 = (R - a_2) \sin\left(\frac{a_1}{R}\right), \quad x_2 = R - (R - a_2) \cos\left(\frac{a_1}{R}\right)$$

R is the radius of curvature of the neutral axis ($a_2 = 0$). This is the exact finite deformation of a beam bent into a circular arc. Fibres at distance a_2

from the neutral axis undergo extension or compression; in the linearised limit $E_{11} \approx -a_2/R$, recovering the classical Euler–Bernoulli bending strain. Students visualise how straight material lines map to circular arcs and compute F and E as functions of position, observing that the strain field is inhomogeneous even though the curvature is uniform.

Physical Example: A metal ruler or a thin steel strip bent by equal and opposite end moments until it takes a circular arc shape.

Case 8 — Radial expansion of a disk

Expressed in Cartesian material coordinates with $r_0 = \sqrt{a_1^2 + a_2^2}$:

$$x_1 = f(r_0) \frac{a_1}{r_0}, \quad x_2 = f(r_0) \frac{a_2}{r_0}$$

where $f(r_0)$ is a prescribed radial mapping (e.g. $f = \lambda r_0$ for uniform radial stretch). The deformation gradient F is most naturally interpreted in polar coordinates; students benefit from computing F in Cartesian coordinates first and then transforming, providing a natural lead-in to curvilinear coordinates and the strain transformation exercise (Task 5).

Physical Example: A gas turbine with its blades rotating at a high speed, expanding outwards.

Case 9 — Tapered uniaxial stretch

$$x_1 = \lambda(a_2) a_1, \quad x_2 = a_2$$

where $\lambda(a_2)$ varies with the transverse coordinate, for example $\lambda = 1 + \varepsilon a_2$. Models a body whose cross-section varies along one direction, such as a tapered bar. The component $F_{21} = \lambda'(a_2) a_1$ introduces a shear term that grows linearly with distance from the axis — a non-obvious result that regularly surprises students encountering inhomogeneous deformations for the first time.

Physical Example: A flat tensile specimen (a dogbone coupon) loaded along its axis. In the gauge section the cross-sectional width varies smoothly with the transverse coordinate, so the local axial stretch $\lambda(a_2)$ is not uniform across the width. The shear component $F_{21} = \lambda'(a_2) a_1$ that grows with distance from the centreline.

Semi-Homogeneous Deformations Involving Waveforms

The displacement field in each case is a waveform modulated over the domain. F varies smoothly from point to point, producing alternating bands of tension, compression and shear that closely resemble photoelastic fringe patterns and DIC full-field strain measurements. These cases are the most visually compelling and most directly connected to experimental observation.

Case 10 — Sinusoidal longitudinal wave

$$x_1 = a_1 + A \sin\left(\frac{2\pi a_1}{L}\right), \quad x_2 = a_2$$

A is the amplitude, L the wavelength. The only non-trivial component of F is $F_{11} = 1 + (2\pi A/L) \cos(2\pi a_1/L)$; all others are trivial. The body alternately compresses and extends along $\mathbf{e}_1$ with spatial period L . Students locate the material points where $J = F_{11} = 0$, giving the condition $2\pi A/L = 1$ for the onset of self-intersection — a physical interpretation of the admissibility requirement $J > 0$.

Physical Example: A one-dimensional compressive ultrasonic wave propagating through a metal bar. The instantaneous displacement pattern at a fixed time is a sinusoidal modulation of the axial position of each material plane. Regions where material planes are crowding together are in compression ($F_{11} < 1$) and regions where they are spreading apart are in tension ($F_{11} > 1$). The condition $J = 0$ corresponds to the wave amplitude at which adjacent material planes would collide — the onset of shock formation.

Case 11 — Sinusoidal transverse wave

$$x_1 = a_1 + A \sin\left(\frac{2\pi a_2}{L}\right), \quad x_2 = a_2$$

The non-trivial gradient component is $F_{12} = (2\pi A/L) \cos(2\pi a_2/L)$ — a spatially varying shear gradient. A rectangular grid of material lines deforms into a wavy pattern; the distortion of the vertical material lines is directly proportional to the local shear strain, making the connection to photoelastic fringe patterns immediate. This is arguably the most visually instructive case in the entire catalogue.

Physical Example: A transverse seismic S-wave propagating horizontally through the ground, or equivalently the snapshot of a vibrating string at a fixed instant. Each horizontal layer of material is displaced laterally by an amount that varies sinusoidally with depth, producing a spatially oscillating

shear strain field. The fringe pattern produced in a photoelastic model under this deformation is visually almost identical to the actual isochromatic fringes observed in loaded transparent specimens, which is why this case makes the connection to photoelasticity so direct.

Case 12 — Biaxial sinusoidal deformation

$$x_1 = a_1 + A \sin\left(\frac{2\pi a_2}{L}\right), \quad x_2 = a_2 + B \sin\left(\frac{2\pi a_1}{L}\right)$$

F is fully populated and varies over the entire domain. The strain field exhibits a checkerboard-like pattern of alternating shear and normal strain concentrations. Visually compelling and a natural analogue of DIC-measured full-field strain maps on specimens with surface texture; students practice identifying strain concentrations by eye before computing (Task 6).

Physical Example: A prominent physical example of biaxial sinusoidal deformation is the formation of surface wrinkles in thin-film structures attached to compliant substrates. When a thin film bonded to a soft substrate is subjected to in-plane biaxial compression, it often develops sinusoidal waves (one-dimensional instability mode) or a square-checkerboard array as an initial instability pattern before potentially transforming into more localized folds or ridges.

Case 13 — Localised necking / shear band

$$x_1 = a_1 + A e^{-a_2^2/w^2} a_1, \quad x_2 = a_2$$

w is the bandwidth parameter. The stretch in the $\mathbf{e}_1$ direction is concentrated in a band centred on $a_2 = 0$ and decays away from it with a Gaussian envelope. Models the onset of a shear band or a neck in a tensile specimen. Students observe that E_{11} is large near the band and negligible away from it, and that the transition zone width is controlled by w — a natural introduction to the concept of a localisation length scale.

Physical Example: The neck that forms in a ductile metal tensile specimen just before fracture — the classic Considère instability. The gauge section develops a localised reduction in cross-sectional area; material inside the neck is stretching rapidly while material outside is barely deforming. The Gaussian envelope $e^{-a_2^2/w^2}$ captures the smooth transition between the deforming zone and the essentially rigid shoulders on either side, and the bandwidth w maps directly onto the observable neck width.

Case 14 — Surface wrinkling

$$x_1 = a_1, \quad x_2 = a_2 + A \sin\left(\frac{2\pi a_1}{L}\right) g(a_2)$$

where $g(a_2)$ is an amplitude envelope: $g = 1$ for uniform wrinkling, $g = e^{-a_2/h}$ for surface-confined wrinkling of penetration depth h . Models buckling of a thin film on a substrate or wrinkling of a compressed membrane. The component $F_{21} = (2\pi A/L) \cos(2\pi a_1/L) g(a_2)$ oscillates in a_1 and decays in a_2 , producing wrinkle-like strain patterns visible in stretched polymer sheets and biological membranes.

Physical Example: The skin of a raisin drying from a grape, the surface of an ageing apple, or more precisely in engineering, the wrinkling of a thin metal or polymer film bonded to a much softer substrate and subjected to compression. The film buckles into a sinusoidal pattern whose wavelength and penetration depth into the substrate are controlled by the stiffness mismatch — exactly the role played by L and h in the mapping. The same geometry describes the mucosal folds of the gut wall and the cortical folding patterns of the brain, making this case a natural bridge toward biomechanics.

(§5.3) Strain/Displacement Relations (Not Included in BME252)

- The finite strain components, E_{ij} , were derived in terms of the deformation *mapping* functions, χ_i .
- It is also possible to write E_{ij} in terms of the displacements, u_i . For that, we introduce

$$u_l = \chi_l - a_l$$

- Take the derivative of the above with respect to the initial coordinates a_l , and solve for χ_l as

$$\frac{\partial \chi_l}{\partial a_i} = \frac{\partial u_l}{\partial a_i} + \delta_{li}$$

- Substitute this into the previous definition of E_{ij}

$$E_{ij} = \frac{1}{2} \left[\delta_{lm} \left(\frac{\partial u_l}{\partial a_i} + \delta_{li} \right) \left(\frac{\partial u_m}{\partial a_j} + \delta_{mj} \right) - \delta_{ij} \right] \quad (7)$$

$$= \frac{1}{2} \left[\frac{\partial u_j}{\partial a_i} + \frac{\partial u_i}{\partial a_j} + \frac{\partial u_l}{\partial a_i} \frac{\partial u_l}{\partial a_j} \right] \quad (8)$$

Unabridged (Classical) Notation

- To retrieve the expression of finite strain in the classical notation, we replace
 - x_1, x_2, x_3 with x, y, z ,
 - a_1, a_2, a_3 with a, b, c ,
 - u_1, u_2, u_3 with u, v, w .
- This yields

$$E_{aa} = \frac{\partial u}{\partial a} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial a} \right)^2 + \left(\frac{\partial v}{\partial a} \right)^2 + \left(\frac{\partial w}{\partial a} \right)^2 \right]$$

- and

$$E_{ab} = \frac{1}{2} \left[\frac{\partial u}{\partial b} + \frac{\partial v}{\partial a} + \left(\frac{\partial u}{\partial a} \frac{\partial u}{\partial b} + \frac{\partial v}{\partial a} \frac{\partial v}{\partial b} + \frac{\partial w}{\partial a} \frac{\partial w}{\partial b} \right) \right]$$

Infinitesimal Motion and Strain

- When the displacements $u_i \ll b$, (they are very small in comparison to the characteristic dimension of the body $\mathcal{B}$),
- Then the difference between the initial and deformed coordinated vanish and we may use

$$\begin{aligned} - a &\rightarrow x \\ - b &\rightarrow y \\ - c &\rightarrow z \end{aligned}$$

- and the second order terms such as $\frac{\partial w}{\partial a} \frac{\partial w}{\partial b}$ may be neglected
- Then the finite strain tensor reduces to the **engineering** or **small** strain tensor, $E_{ij} \rightarrow \epsilon_{ij}$ given by

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right)$$

Small Strains in Classical Notation

When the small strains are expressed in classical notation

- x_1, x_2, x_3 with x, y, z ,
- u_1, u_2, u_3 with u, v, w .

$$\begin{aligned} e_{xx} &= \frac{\partial u}{\partial x}, & e_{xy} &= \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = e_{yx}, \\ e_{yy} &= \frac{\partial v}{\partial y}, & e_{xz} &= \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) = e_{zx}, \\ e_{zz} &= \frac{\partial w}{\partial z}, & e_{yz} &= \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) = e_{zy}. \end{aligned} \tag{5.3-7}$$